Fiscal Year:	FY 2006	**Task Last Updated:**	FY 11/09/2005
PI Name:	Judge, Andrew Ph.D.		
Project Title:	The Use of Aspirin and Other NSAIDs to Ameliorate Muscle Atropy Due to Simulated Weightlessness		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Muscle Alterations and Atrophy Team		
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Space Biology Element:	None		
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Zip Code:	02215	**Congressional District:**	8
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Contact Monitor:		**Contact Phone:**	
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Flight Assignment:	End date of 11/30/2006 changed to new end date of 9/30/2006		
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COI Name (Institution):	Kandarian, Susan (MENTOR/Boston University)		
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	POSTDOCTORAL FELLOWSHIP The overall objective of the present application is to test the ability of aspirin and other non-steroidal anti-inflammatory drugs (NSAID) to inhibit skeletal muscle atrophy, weakness, and changes in gene expression due to simulated weightlessness in rats using the hind limb suspension model. The specific aims for this proposal are to: 1) determine if high dose aspirin treatment inhibits: soleus and plantaris fiber atrophy, deficits in maximum force, and the activation of NF-kappa B seen after 7 days of hindlimb unloading. The inhibition of cyclooxygenase activity by NSAID treatment will also be assessed in muscles; 2) determine if curcumin or naproxen treatment inhibits: soleus and plantaris fiber atrophy, deficits in maximum force, and NF-kappa B or cyclooxygenase activity due to 7 days of unloading; 3) determine if the expression of genes known to be upregulated with 7 days of unloading are “normalized” by treatment		

Task Description:	with any of the NSAIDs. To determine if curcumin inhibits NF-kappa B activity, C2C12 mouse myotubes were treated with 10 ngrams/ml TNF-alpha (a potent activator of NF-kappa B) for 3 hours either in the presence or absence of 200 mM curcumin. While TNF-alpha caused a 9-fold increase in NF-kappa B activity, curcumin completely abolished this increase. These results suggest that curcumin may be an effective countermeasure. However, in hind limb unloaded animals injection of curcumin did not attenuate the increase in NF-kapp B activity, nor did it attenuate the unloading-induced muscle atrophy. While disappointing, this is in agreement with a very recent study showing dietary curcumin does not attenuate unloading-induced NF-kappa B activity in skeletal muscle or unloading-induced skeletal muscle dysfunction or atrophy (Farid et al. Effects of dietary curcumin or N-acetylcysteine on NF-?B activity and contractile performance in ambulatory and unloaded murine soleus. Nutr Metab (Lond). 2005; 2: 20). The ineffectiveness of curcumin in vivo may be due to the low bioavailability. To determine if aspirin inhibits NF-kappa B activity, C2C12 mouse myotubes were treated with 10 ngrams/ml TNF-alpha for 3 hours either in the presence or absence of 20mM aspirin. TNF-alpha caused an 11-fold increase in NF-kappa B activity and aspirin abolished this increase. In unloaded animals, injection of aspirin did not attenuate the increase in NF-kappa B activity, but it did attenuate the unloading-induced muscle atrophy. Therefore, this dose of aspirin may be working by the inhibition of other molecules, which we have yet to test. Although the NSAIDs outlined above have pleiotropic effects, both aspirin and curcumin inhibit NF-kappa B activity by inhibiting I kappa B kinase. Therefore to gain further insight into other components of the NF-kappa B pathway that are necessary for unloading-induced atrophy a fourth and fifth aim were added using genetic approaches to focus on the I kappa B alpha protein (which is downstream of I kappa B kinase) and the c-rel gene. By determining the involvement of particular proteins and genes in the atrophy process, specific pharmacological countermeasures may be tested in subsequent studies. Aim 4: To determine if I kappa B alpha is required for unloading-induced NF-?B activity, muscle atrophy, and upregulation of atrophy-related genes. For this aim an I kappa B alpha dominant negative plasmid (known as a “superrepressor”) was injected into the soleus muscle of rats. This superrepressor is resistant to phosphorylation by I kappa B kinase and therefore resistant to ubiquitination and subsequent degradation. NF-kappa B activity was increased 5-fold with hind limb unloading, but completely abolished in the unloaded muscles injected with the superrepressor. Furthermore, soleus muscle fiber cross sectional area was decreased by 40% following 7 days unloading but was attenuated by 40% in the unloaded muscles injected with the superrepressor. In addition the increase in gene expression of atrogin-1/MAFbx, Cathepsin L, Nedd4, IEX-1, 4E-BP1 and FOXO3a with unloading was significantly attenuated in the unloaded muscles injected with the superrepressor. Aim 5: To determine if c-rel is required for unloading-induced NF-kappa B activity and muscle atrophy. For this aim we unloaded wild type (WT) and c-rel knockout (-/-) mice. NF-kappa B activity was increased 8-fold, with unloading, in the soleus muscle of WT mice, and remained equally elevated with unloading in the c-rel -/- mice. Moreover, muscle atrophy was the same in WT unloaded and c-rel -/- unloaded. This eliminates c-rel as a contributor to the atrophy process and, therefore, as a therapeutic target. Ongoing experiments will test whether naproxen and possibly celecoxib inhibit skeletal muscle atrophy, deficits in maximum force, and NF-kappa B or cyclooxygenase activity due to 7 days of unloading.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	By identifying specific proteins and genes involved in skeletal muscle atrophy and testing nutritional and/or pharmacological countermeasures to them we will be in a much better position to combat the deleterious changes in muscle function due to any type of disuse.
Task Progress:	POSTDOCTORAL FELLOW. To determine if curcumin inhibits NF-kappa B activity, C2C12 mouse myotubes were treated with 10 ng/ml TNF-alpha (a potent activator of NF-kappa B) for 3 hours either in the presence or absence of 200mM curcumin. While TNF-alpha caused a 9-fold increase in NF-kappa B activity, curcumin completely abolished this increase. These results suggest that curcumin may be an effective countermeasure. However, in hind limb unloaded animals injection of curcumin did not attenuate the increase in NF-kappa B activity, nor did it attenuate the unloading-induced muscle atrophy. The ineffectiveness of curcumin in vivo may be due to the low bioavailability. To determine if aspirin inhibits NF-kappa B activity, C2C12 mouse myotubes were treated with 10 ng/ml TNF-alpha for 3 hours either in the presence or absence of 20mM aspirin. TNF-alpha caused an 11-fold increase in NF-kappa B activity and aspirin abolished this increase. In unloaded animals, injection of aspirin did not attenuate the increase in NF-kappa B activity, but it did attenuate the unloading-induced muscle atrophy. Therefore, this dose of aspirin may be working by the inhibition of other molecules, which we have yet to test. To determine if I kappa B alpha is required for unloading-induced NF-kappa B activity, muscle atrophy, and upregulation of atrophy-related genes we injected an I kappa B alpha dominant negative plasmid (known as a “superrepressor”) into the soleus muscle of rats. This superrepressor is resistant to phosphorylation by I kappa B kinase and therefore resistant to ubiquitination and subsequent degradation. In this study NF-kappa B activity was increased 5-fold in the soleus muscle with unloading, but completely abolished in the unloaded muscles injected with the superrepressor. Furthermore, soleus muscle fiber cross sectional area was decreased by 40% following 7 days unloading but was attenuated by 40% in the unloaded muscles injected with the superrepressor. In addition the increase in gene expression of atrogin-1/MAFbx, Cathepsin L, Nedd4, IEX-1, 4E-BP1 and FOXO3a with unloading was significantly attenuated in the unloaded muscles injected with the superrepressor. To determine if c-rel is required for unloading-induced NF-kappa B activity and muscle atrophy we unloaded wild type (WT) and c-rel knockout (-/-) mice. NF-kappa B activity was increased 8-fold with unloading in the soleus muscle of WT mice, and remained equally elevated with unloading in the c-rel -/- mice. Furthermore muscle atrophy was the same in WT unloaded and c-rel -/- unloaded. This eliminated c-rel as a contributor to the atrophy process and, therefore, as a therapeutic target.
Bibliography Type:	Description: (Last Updated: 05/07/2007)
Awards	Judge A. "Young Investigator Award, FASEB Summer Research Conference on Skeletal Muscle and Stem Cells, July 2005." Jul-2005
Presentation	Judge, A. and S. C. Kandarian "NF-KAPPA B AND MUSCLE ATROPHY" N/A Jul-2005

Presentation	Judge, A., A. Koncarevic, and S. C. Kandarian "INHIBITION OF I KAPPA B ALPHA DEGRADATION PREVENTS NF-KAPPA B ACTIVATION AND ATTENUATES FIBER ATROPHY AND GENE EXPRESSION ASSOCIATED WITH UNLOADING ATROPHY." N/A Jun-2005